

Incident Investigation Report: MN-DW-02 Gas Influx at 4,195 m MD (illustrative)

Document ID: INC-MN-DW-02-KICK-4195 | Type: INCIDENT | Well: MN-DW-02 | Date: 2024-10-18

1. Event Identification

- Well**: MN-DW-02
- Date & Time**: 14-Oct-2024, 03:42 IST
- Depth**: 4,195 m MD
- Event Type**: Hydrocarbon Influx / Well Control Event

2. Description of Influx

While drilling 12-1/4 in section with 11.15 ppg (1.34 SG) SOBM, the bit penetrated top Miocene turbidite channel sand at 4,195 m MD. Mud log recorded instantaneous gas peak of 18.4% (methane 88%, ethane 7%, propane 5%).

A 12 bbl pit gain was confirmed on the active PVT system.

Well was shut in via soft shut-in protocol:

- SIDPP: 236 psi
- SICP: 310 psi

Calculated formation pore pressure: **11.48 ppg (1.38 SG)**.

3. Root Cause Analysis

- Failure to recognize transition zone geophysical indicators (sonic slowness excursion $> 12 \text{ } \mu\text{s/ft}$).
- Delayed mud weight adjustment due to missing pre-emptive weight-up protocol.
- Influx killed successfully with 11.65 ppg (1.40 SG) mud. Total NPT: 36.5 hours.